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ICS 2203 WEB APPLICATION DEVELOPMENT 1

LAST REVISION ON October 3, 2013

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 6

Introduction to HTML

Learning outcomes

Upon completing this topic, you should be able to:

By the end of this class the student should be able to:

1. Identify different web technologies used in web programming.
2. Employ html technology in creating simple web pages.
3. Generate web pages through the use of a web development tool.



6.1. Introduction to HTML

HTML is a document-layout and hyperlink-specification language. It defines the syntax and placement of special, embedded tags usually NOT displayed by the browser, but tell it how to display the contents of the document, including text, images, and other support media. The language also tells you how to make a document interactive through special hypertext links, which connect your document with other documents - on either your computer or someone else's, as well as with other Internet resources, like FTP. The basic syntax and semantics of HTML are defined in the HTML standard. The HTML language has continuously been changing since its inception to incorporate the changes in technology. Currently the most current version of HTML is HTML 5



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Browser developers rely upon the HTML standard to program the software that formats and displays common HTML documents. Authors use the standard to make sure they are writing effective, correct HTML documents. Nonetheless, commercial forces have pushed developers to add into their browsers - Netscape Navigator and Internet Explorer, in particular - non-standard extensions meant to improve the language. Many times, these extensions are implementations of future standards still under debate. Extensions can foretell future standards because so many people use them.

Let us begin looking at the HTML language by first of all understanding the structure



6.2. Structure of an HTML Document

HTML documents consist of text, which defines the content of the document, and tags, which define the structure and appearance of the document. The structure of an HTML document is simple, consisting of an outer `<html>` tag enclosing the document head and body:

- `<html>`
- `<head>`
- `<title>Barebones HTML Document</title>`
- `</head>`
- `<body>`
- This illustrates, in a very *simple* way,
- the basic structure of an HTML document.



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- `</body>`
- `</html>`

Each document has a head and a body, delimited by the `<head>` and `<body>` tags. The head is where you give your document a title and where you indicate other parameters the browser may use when displaying the document. The body is where you put the actual contents of the document. This includes the text for display and document control markers (tags) that advise the browser how to display the text. Tags also reference special-effects files, including graphics and sound, and indicate the hot spots (hyperlinks and anchors) that link your document to other documents.



6.3. Syntax of a Tag

Every tag consists of a tag name, sometimes followed by an optional list of tag attributes, all placed between opening and closing brackets (< and >). The simplest tag is nothing more than a name appropriately enclosed in brackets, such as <head> and <i>. More complicated tags contain one or more attributes, which specify or modify the behavior of the tag.

According to the HTML standard, tag and attribute names are not case-sensitive. There's no difference in effect between <head>, <Head>, <HEAD>, or even <HeaD>; they are all equivalent. With XHTML, case is important: all current standard tag and attribute names are in lowercase. Tag attributes, if any, belong after the tag name, each separated by one or more tab, space, or return characters. Their order of appear-



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ance is not important. A tag attribute's value, if any, follows an equal sign (=) after the attribute name. You may include spaces around the equal sign, so that `width=6`, `width = 6`, `width =6`, and `width= 6` all mean the same. For readability, however, we prefer not to include spaces. That way, it's easier to pick out an attribute/value pair from a crowd of pairs in a lengthy tag.

6.4. Sample Tags

Here are some tags with attributes:

- `<a href="http://www.oreilly.com/catalog.html">`
- `<ul compact>` `<ul compact="compact">`
- `<input type=text name=filename size=24 maxlength=80>`
- `<link title="Table of Contents">`



6.5. Starting and Ending Tags

We alluded earlier to the fact that most tags have a beginning and an end and affect the portion of content between them. That enclosed segment may be large or small, from a single text character, syllable, or word, to the `<html>` tag that bounds the entire document. The starting component of any tag is the tag name and its attributes, if any. The corresponding ending tag is the tag name alone, preceded by a slash. Ending tags have no attributes.

6.6. HTML Document Elements

Every HTML document should conform to the HTML SGML DTD, the formal Document Type Definition that defines the HTML standard. The DTD defines the tags and syntax that are



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used to create an HTML document. You can inform the browser which DTD your document complies with by placing a special SGML (Standard Generalized Markup Language) command in the first line of the document:

```
<!DOCTYPE HTML PUBLIC "-  
//W3C//DTD HTML 4.01//EN">
```

This cryptic message indicates that your document is intended to be compliant with the HTML 4.01 final DTD defined by the World Wide Web Consortium (W3C). Other versions of the DTD define more restricted versions of the HTML standard, and not all browsers support all versions of the HTML DTD. In fact, specifying any other doctype may cause the browser to misinterpret your document when displaying it for the user. We will talk more about DOCTYPE in a later class



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An element refers to both the opening and closing tags plus the text or content being affected. i.e `< I>` this is italics `</I>`

6.7. Classification of HTML TAGS and ELEMENTS

HTLM has a vast collection of tags for use that are supported by most browsers. To better Understand these we may classify the tags based on the following:

Presentation Tags – Category of tags that solely affects the appearance (look and feel of a web page) these may be further categorized into physical tags and content based tags.

Layout Tags- these are tags that specifically address layout aspects of the web page. They define the sections, paragraphs and general organization of the web page. Other



Tags- here we have the many other tags that do not necessarily fall in to the two categories we have above but perform other various functions on a web page

6.8. Changing Text Appearance (Presentation Tags)

A number of tags change the appearance of and associate hidden meaning with text. In general, these tags can be grouped into two: content-based styles and physical styles. In addition, the W3C standard for Cascading Style Sheets is now well-supported by the popular browsers, providing another, more comprehensive way for authors to control the look and layout of their document text.



6.8.1. Content-Based Styles

Content-based style tags inform the browser that the enclosed text has a specific meaning, context, or usage. The browser then formats the text in a manner consistent with that meaning, context, or usage.

Because font style is specified via semantic clues, the browser can choose a display style that is appropriate for the user. Since such styles vary by locale, using content-based styles helps ensure that your documents will have meaning to a broader range of readers. This is particularly important when a browser is targeted at blind or handicapped readers whose display options are radically different from conventional text or are extremely limited in some way. The current HTML and XHTML standards do not define a format for each of the content-based styles ex-



cept that they must be rendered in a manner different from the regular text in a document. The standard doesn't even insist that the content-based styles be rendered differently from one another. In practice, you'll find that many of these tags have fairly obvious relationships with conventional print, having similar meanings and rendered styles, and are rendered in the same style and fonts by most browsers.

6.8.2. Physical Styles

We use the word "intent" a lot when we talk about content-based style tags. That's because the meaning conveyed by the tag is more important than the way a browser displays the text. In some cases, however, you might want the text to appear explicitly in some special way – italic or bold, for example – perhaps



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for legal or copyright reasons. In those cases, use a physical style for the text.

While the tendency with other text-processing systems is to control style and appearance explicitly, with HTML or XHTML you should avoid explicit, physical tags except on rare occasions. Provide the browser with as much contextual information as possible. Use the content-based styles. Even though current browsers may do nothing more than display their text in italic or bold, future browsers and various document-generation tools may use the content-based styles in any number of creative ways.





6.8.3. Content-Based Styles

- **Content-Based Styles**

The `<abbr>` tag indicates that the enclosed text is an abbreviated form of a longer word or phrase. The browser might use this information to change the way it renders the enclosed text.

- **The `<acronym>` Tag**

The `<acronym>` tag indicates that the enclosed text is an acronym, an abbreviation formed from the first letter of each word in a name or phrase, such as HTML and IBM. `<acronym>`.

- **The `<cite>` Tag**

The `<cite>` tag usually indicates that the enclosed text is a bibliographic citation like a book or magazine title. By convention,



the citation text is rendered in *italic*. Try:

While kumquats are not mentioned in Melville's `<cite>Moby Dick</cite>`, it is nonetheless apparent that the mighty cetacean represents the bitter "kumquat-ness" within every man. Indeed, when Ahab spears the beast, its flesh is tough, much like the noble fruit.

- The `<code>` Tag

Software code warriors have become accustomed to a special style of text presentation for their source programs. The `<code>` tag is for them. It renders the enclosed text in a monospaced, teletype-style font like Courier.

The array reference `<code>a[i]</code>` is identical to the



pointer reference `<code>*(a+i)</code>`.

- The `<dfn>` Tag

Use `<dfn>` to tag those defining instances of special terms or phrases. It may not result in any formatting changes by the browser. Instead, `<dfn>` might assist in creating a document index or glossary.

Example. For example, use the `<dfn>` tag to introduce a new phrase to the reader: When analyzing annual crop yields, `<dfn>`rind spectroscopy`</dfn>` may prove useful. By comparing the relative levels of saturated hydrocarbons in fruit from adjacent trees, rind spectroscopy has been shown to be 87% effective in predicting an outbreak of trunk dropsy in trees under four years old.



- The `<em>` Tag

The `<em>` tag tells the client browser to present the enclosed text with emphasis. For nearly all browsers, this means the text is rendered in italic. For example, the popular browsers will emphasize by italicizing the words "always" and "never" in the following HTML source:

Kumquat growers must `<em>always</em>` refer to kumquats as "the noble fruit," `<em>never</em>` as just a "fruit."

- The `<kbd>` Tag

As you probably already suspect, it is used to indicate text that is typed on a keyboard. Its enclosed text typically is rendered by the browser in monospaced font style. The `<kbd>` tag is most often used in computer-related documentation and manu-



als, such as in this example:

Example. Type `<kbd>quit</kbd>` to exit the utility, or type `<kbd>menu</kbd>` to return to the main menu.

- **The `<samp>` Tag**

The `<samp>` tag indicates a sequence of literal characters that should have no other interpretation by the user. This tag is most often used when a sequence of characters is taken out of its normal context. For example, the following source:

Example. The `<samp>ae</samp>` character sequence may be converted to the `æ` ligature if desired.

The `<samp>` tag is not used very often. It should be used in those few cases where special emphasis needs to be placed on small character sequences taken out of their normal context.



- The `<strong>` Tag

Like the `<em>` tag, the `<strong>` tag is for emphasizing text, except with more gusto. Browsers typically display the `<strong>` tag differently than the `<em>` tag, usually by making the text bold (versus italic), so that users can distinguish between the two.

Example. For example, in the following text, the emphasized "never" appears in italic with Internet Explorer, while the `<strong>` "forbidden" is rendered in bold characters. One should `<em>`never`</em>` make a disparaging remark about the noble fruit. In particular, mentioning kumquats in conjunction with vulgar phrases is expressly `<strong>`forbidden`</strong>` by the Association by-laws.



- The `<var>` Tag

The `<var>` tag, another computer-documentation trick, indicates a variable name or a user-supplied value. The tag is often used in conjunction with the `<code>` and `<pre>` tags for displaying particular elements of computer programming code samples and the like. `<var>`-tagged text typically is rendered in italics, as shown in The user should type

```
<pre> cp <var>source-file</var>
```

```
<var>dest-file</var> </pre> replacing the <var>  
source-file</var>
```

with the name of the source file, and

```
<var>dest-file</var>
```

with the name of the destination file.



6.8.4. Physical Styles

- The `<b>` Tag

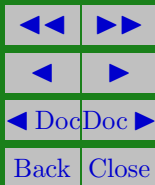
The `<big>` tag makes it easy to increase the size of text. It couldn't be simpler: the browser renders the text between the `<big>` tag and its matching `</big>` ending tag one font size larger than the surrounding text. If that text is already at the largest size, `<big>` has no effect. Even better, you can nest `<big>` tags to enlarge the text. Each `<big>` tag makes the text one size larger, up to a limit of size seven, as defined by the font model. Be careful with your use of the `<big>` tag, though. Because browsers are quite forgiving and try hard to understand a tag, those that don't support `<big>` often interpret it to mean bold.



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- The `<blink>` Tag

Text contained between the `<blink>` tag and its end tag `</blink>` does just that: blink on and off. Netscape for Macintosh, for example, simply and reiteratively reverses the background and foreground colors for the `<blink>` enclosed text. Neither the HTML nor the XHTML standard include `<blink>`; only Netscape Navigator supports it as an extension. We cannot effectively reproduce the animated effect in these static pages, but it is easy to imagine and best left to the imagination, too. That's because blinking text has two primary effects: it gets your reader's attention, and then promptly annoys them to no end. Blinking text should be used sparingly in any context.





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- The `<i>` Tag

The `<i>` tag is like the `<em>` content-based style tag. It and its necessary (`</i>`) end tag tell the browser to render the enclosed text in an italic or oblique typeface. If the typeface is not available to the browser, highlighting, reverse video, or underlining might be used.

- The `<s>` Tag

The `<s>` tag is an abbreviated form of the `<strike>` tag supported by both Internet Explorer and Netscape. It is now a deprecated tag in HTML 4 and XHTML, meanings don't use it; it will eventually go away.



- The `<small>` Tag

The `<small>` tag works just like its `<big>` counterpart (see previous description), except it decreases the size of text instead of increasing it. If the enclosed text is already at the smallest size supported by the font model, `<small>` has no effect. Like `<big>`, you may also nest `<small>` tags to sequentially shrink text. Each `<small>` tag makes the text one size smaller than the containing `<small>` tag, down to a limit of size one.

- The `<strike>` Tag

Most browsers will put a line through ("strike through") text that appears inside the `<strike>` tag and its `</strike>` end tag. Presumably, it is an editing markup that tells the reader to ignore the text passage, reminiscent of the days before typewriter



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correction tape. You'll rarely, if ever, see the tag in use today, and probably never will: it is deprecated in HTML 4 and XHTML, just one version away from complete elimination from the standard.

- **The `<sub>` Tag**

The text contained between the `<sub>` tag and its `</sub>` end tag gets displayed half a character's height lower, but in the same font and size as the current text flow. Both `<sub>` and its `<sup>` counterpart are useful for math equations and in scientific notation, as well as with chemical formulæ.

- **The `<sup>` Tag**

The `<sup>` tag and its `</sup>` end tag superscripts the enclosed text; it gets displayed half a character's height higher,



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but in the same font and size as the current text flow. This tag is useful for adding footnotes to your documents, along with exponential values in equations. In combination with the `<a>` tag, you can create nice, hyperlinked footnotes: The larval quat weevil

```
<a href="footnotes.html#note74">
```

```
<sup><small>74</small></sup></a> is a
```

- **The `<tt>` Tag**

In a manner like the `<code>` and `<kbd>` tags, the `<tt>` tag and necessary `</tt>` end tag direct the browser to display the enclosed text in a monospaced typeface. For those browsers that already use a monospaced typeface, this tag may make no discernible change in the presentation of the text.



- The `<u>` Tag

This tag tells the browser to underline the text contained between the `<u>` and the corresponding `</u>` tag. The underlining technique is simplistic, drawing the line under spaces and punctuation as well as the text. This tag is deprecated in HTML 4 and XHTML and will be eliminated in the next version of the standard. The same effect can be achieved by using style sheets, covered in

Example. HTML Example;

Type in the following in an editor and save as an .html file then answer the questions that follow.

```
<html>< head> < title> a HTML tutorial
```

```
</title> </head><body>
```

```
<p>In general tags are what actually make up the markup in
```



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html. We have different types of tags that perform the functions of marking up a document to give it a good presentation to its viewers.

`< /b>` in this tutorial you are required to note how these tags have been used

`<p>` we can classify some of the major tags into two categories namely:

`<ol> <li>` presentation or physical style elements `</li>`

`<li>` context based or emphasis elements `</li></ol>`

`<h1 align = centered>`Presentation Elements`</h1>`

These are concerned with the appearance of the text in general. The bolding italicizing, underlining and basically formatting that has to do with how the text appears on a browser. i.e.



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- `<i>` this is italics`</i>`
- `<b>`this is bolded`</b>`
- `<big>` this is big`</big>`

`<blink>`this causes the text to blink on some browsers`</blink>`
`<s>`this is strike through`</s>`
`<small>` this is small`</small>`
`<sup>`this is superscript.`</sup>`
`<sub>` this is sub script`</sub>`
`<tt>` this is mono spaced font`</tt>`
`<u>` this is underlined`</u>`
Explicitly `<b>`boldfaced`</b>`, `<i>`italicized`</i>`, or
`<tt>`teletype-style`</tt>` text should be used
`<big>``<big>`sparingly`</big>``</big>`.

Otherwise, drink

`<strike>`lots`</strike>` 1x10`<sup>`6`</sup>`
drops of H`<sub>``<small>``<small>`2
`</small>``</small>``</sub>`O.



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<p>Any physical style tag may contain any item allowed in text, including conventional text, anchors, images, and line breaks. You also can combine physical style tags with other content-based ones.

<h1 align = centered> Content Based tags< /h1>

these include the following:

 <abbr> the abbreviation

</ abbr>

<acronym>the acronym</acronym>

<cite>the cite</cite>< br>

for example While kumquats are not mentioned in

Melville's<cite>Moby Dick</cite> ,

it is nonetheless apparent that the mighty cetacean represents the bitter "kumquat-ness" within every man. Indeed, when



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Ahab spears the beast, its flesh is tough, much like the noble fruit.

`the code` for example The array reference `a[i]` is identical to the pointer reference `*(a+i)`.

for definition

i.e. When analyzing annual crop yields, rind spectroscopy may prove useful. By comparing the relative levels of saturated hydrocarbons in fruit from adjacent trees, rind spectroscopy has been shown to be 87% effective in predicting an outbreak of trunk dropsy in trees under four years old.

the emphasis tag

the keyboard tag

used when a sequence of characters is taken out of its normal context

i.e. The `ae` character sequence may be



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converted to the æ ligature if desired
and finally the strong also for more emphasis

<p> the context tag unlike the physical ones carries meaning alongside the formatting. This is especially useful for browsers that have screen readers for eye disabled persons to actually get the meaning in context. For example the strong tag when a browser encounters it will pronounce it ‘with emphasis when using a screen reader.

<h1 align = centerd> Other Tags and elements affecting text< /h1>
Unless the base font size was reset above,

Internet Explorer renders this part in font-size 3.<basefont size=7> This text should be rather large (size 7). <basefont size=6> Oh,<basefont size=4> no!<basefont size=2> I’m<basefont size=1> shrinking!</basefont>
 Ahhhh, back to normal.



<p>

The ghost moaned,

”oooooo

oooooooo.”

This text is in the default font. But, heaven only knows what font face is this one?</body> </html>

6.9. Changing the Page Layout (Layout Tags)

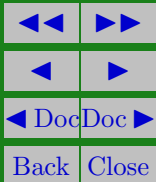
6.9.1. HTML Frames

HTML frames allow authors to present documents in multiple views, which may be independent windows or sub windows. Multiple views offer designers a way to keep certain information visible, while other views are scrolled or replaced. For exam-



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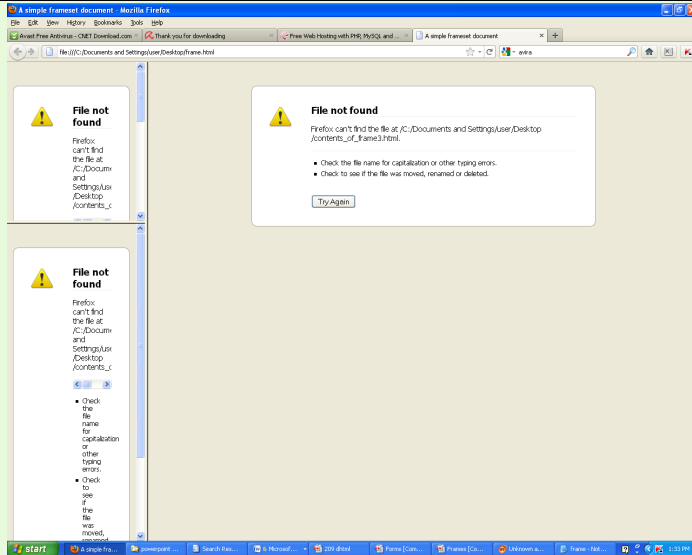
ple, within the same window, one frame might display a static banner, a second a navigation menu, and a third the main document that can be scrolled through or replaced by navigating in the second frame.



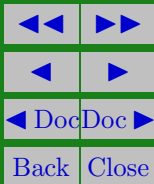
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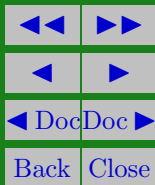
The Fig Above shows a frame which has three different windows





6.9.2. Creating a Frame on a web Page

Frames are created using the Frameset tag together with setting the rows and columns attributes of the tag. Frames are created left-to-right for columns and top-to-bottom for rows. When both attributes are specified, views are created left-to-right in the top row, left-to-right in the second row, etc. Setting the rows attribute defines the number of horizontal subspaces in a frameset. Setting the cols attribute defines the number of vertical subspaces. Both attributes may be set simultaneously to create a grid. If the rows attribute is not set, each column extends the entire length of the page. If the cols attribute is not set, each row extends the entire width of the page. If neither attribute is set, the frame takes up exactly the size of the page.





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Example . In the example below, the screen is divided vertically in two (i.e., creates a top half and a bottom half).

<FRAMESET rows="50%, 50 %"> ...the rest of the definition... </FRAMESET> The values for the frameset row or column may be integers (100), percentages(30% or wildcards(2*).

NOTE: Frameset documents do not have the body section. This is because frames will have multiple web pages in one window.

Example . A Frameset Document

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01  
Frameset//EN" "http://www.w3.org/TR/html4/frameset.dtd">
```

```
<HTML> <HEAD> <TITLE>
```

A simple frameset document



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```
</TITLE> </HEAD> <FRAMESET cols="20%, 80%">
<FRAMESET rows="100, 200"> <FRAME src="contents_of_frame1.html">
<FRAME src="contents_of_frame2.gif"> </FRAMESET>
<FRAME src="contents_of_frame3.html"> <NOFRAMES>
<P>This frameset document contains: <UL> <LI><A href="contents_of_frame1.html">
Some neat contents</A> <LI><IMG src="contents_of_frame2.gif">
alt="A neat image">
<LI><A href="contents_of_frame3.html">Some other neat
contents</A>
</UL> </NOFRAMES> </FRAMESET> </HTML>
```

6.9.3. Other Examples

- The next example creates three columns: the second has a fixed width of 250 pixels (useful, for example, to hold an

image with a known size). The first receives 25% of the remaining space and the third 75% of the remaining space.

```
<FRAMESET cols="1*,250,3*"> ...the rest of the definition...  
</FRAMESET>
```

- The next example creates a 2x3 grid of subspaces.

```
<FRAMESET rows="30%,70%" cols="33%,34%,33%"> ...the rest of the definition... </FRAMESET>
```
- For the next example, suppose the browser window is currently 1000 pixels high. The first view is allotted 30% of the total height (300 pixels). The second view is specified to be exactly 400 pixels high. This leaves 300 pixels to be divided between the other two frames. The fourth frame's height is specified as "2*", so it is twice as high as the third frame, whose height is only "*" (equivalent to 1*). There-





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fore the third frame will be 100 pixels high and the fourth will be 200 pixels high.

<FRAMESET rows="30%,400,*,2*"> ...the rest of the definition... </FRAMESET>

6.9.4. HTML Forms

A HTML form is a section of a document containing normal content, markup, special elements called controls (checkboxes, radio buttons, menus, etc.), and labels on those controls. Users generally "complete" a form by modifying its controls (entering text, selecting menu items, etc.), before submitting the form to an agent for processing (e.g., to a Web server, to a mail server, etc.).

HTML Forms enhance interactivity of web documents by



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allowing creators or owners get feedback. Bellow is a simple form that includes labels, radio buttons, and push buttons (reset the form or submit it):

```
<FORM action="http://somesite.com/prog/adduser" method="p
<P> <LABEL for="firstname">First name: </LABEL>
<INPUT type="text" id="firstname"><BR>
<LABEL for="lastname">Last name: </LABEL>
<INPUT type="text" id="lastname"><BR>
<LABEL for="email">email: </LABEL>
<INPUT type="text" id="email"><BR>
<INPUT type="radio" name="sex" value="Male"> Male<BR>
<INPUT type="radio" name="sex" value="Female">
Female<BR> <INPUT
type="submit" value="Send">
```





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```
<INPUT type="reset"> </P> </FORM>
```

```
<html>
```

```
<body>
```

```
preceding content...
```

```
<a name="Stews">
```

```
<h3>Kumquat Stew Recipes</h3>
```

```
</a>
```

In the same or another

For more information on kumquats, visit our

```
<a href="http://www.kumquat.com/archive.html">
```

Kumquat Archive,

and perhaps try one or two of our

```
<a href="http://www.kumquat.com/archive.html#Stews">
```



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Kumquat Stew Recipes.

```
<p><IMG SRC = "angel.jpeg">
```

```
<ul>
```

```
<li><a href="default.html">Home</a></li>
```

```
<li><a href="about/about.htm">About</a></li>
```

```
<li><a href="news/news.htm">News</a></li>
```

```
<li><a href="calendar/calendar.htm">Calendar</a></li>
```

```
<li><a href="photo_gallery/photo_gallery.htm">Photo  
Gallery</a></li>
```

```
<li><a href="information_links/information_links.htm">  
Links</a></li>
```

```
<li><a href="faq/faq.htm">FAQ</a></li>
```

```
<li><a href="contact/contact.htm">Contact</a></li>
```



```
</ul>
<table border cellpadding=5>
<caption align=bottom>
Kumquat versus a poked eye, by gender</caption>
<tr>
<td colspan=2 rowspan=2></td>
<th colspan=2 align=center>Preference</th>
</tr>
<tr>
<th>Eating Kumquats</th>
<th>Poke In The Eye</th>
</tr>
<tr align=center>
```



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```
<th rowspan=2>Gender</th>
<th>Male</th>
<td>73%</td>
<td>27%</td>
</tr>
<tr align=center>
<th>Female</th>
<td>16%</td>
<td>84%</td>
</tr>
</table>
</body>
```



</html>

Figure showing the browser output of the Form created
The Form Element is used to define a HTML Form. It has



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two attributes that are required.

- **Attribute definitions**

- action = uri [CT]
- This attribute specifies a form processing agent. User agent behavior for a value other than an HTTP URI is undefined.
- method = get|post [CI]
- This attribute specifies which HTTP method will be used to submit the form data set. Possible (case-insensitive) values are "get" (the default) and "post". See the section on form submission
- <FRAMESET rows="30%,70%" cols="33%,34%,33%">

on for usage information.



6.9.5. When to use post or get

- Which one to use if your form-processing server supports both the POST and GET methods? Here are some rules of thumb:
- For best form-transmission performance, send small forms with a few short fields via the GET method.
- Because some server operating systems limit the number and length of command-line arguments that can be passed to an application at once, use the POST method to send forms that have many fields or that have long text fields.
- If you are inexperienced in writing server-side form-processing applications, choose GET. The extra steps involved in reading and decoding POST-style transmitted parameters,



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while not too difficult, may be more than you are willing to tackle.

- If security is an issue, choose POST. GET places the form parameters directly in the application URL where they easily can be captured by network sniffers or extracted from a server log file. If the parameters contain sensitive information like credit card numbers, you may be compromising your users without their knowledge. While POST applications are not without their security holes, they can at least take advantage of encryption when transmitting the parameters as a separate transaction with the server.
- If you want to invoke the server-side application outside the realm of a form, including passing it parameters, use
- GET because it lets you include form-like parameters as



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part of a URL. POST-style applications, on the other hand, expect an extra transmission from the browser after the URL, something you can't do as part of a conventional `<a>` tag.

Exercise. Try this one out

```
<form method=POST action="
http://www.kumquat.com/demo">
```

Name:

```
<input type=text name=name size=32 maxlength=80>
```

```
<p>
```

Sex:

```
<input type=radio name=sex value="M">
```

Male

```
<input type=radio name=sex value="F">
```

 Female

```
<p>
```

Income:

```
<select name=income size=1>
```



<option>Under \$25,000

<option>\$25,001 to \$50,000 <option>\$50,001 and higher

</select>

<p>

<input type=submit>

</form>

- **The Input Tag**

Use the <input> tag to define any one of a number of common form "controls," as they are called in the HTML 4 and XHTML standards, including text fields, multiple-choice lists, clickable images, and submission buttons. Although there are many attributes for the <input> tag, only the type and name attributes are required for each element (only type for a submission or re-



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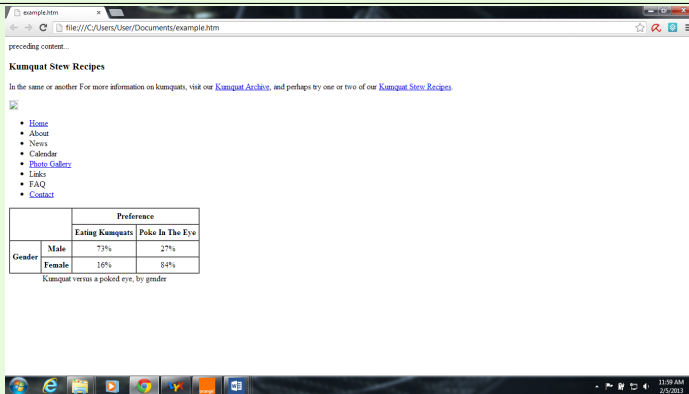
set button; see following explanation). And as we describe in detail later, each type of input control uses only a subset of the allowed attributes. Additional `<input>` attributes may be required based upon which type of form element you . The Type attribute define the type of control to be inserted i.e. if type = checkbox then a checkbox control will be created and so on.

To include HTML TABLES **Example** . Consider the Browser output below.





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Required: Write the html code responsible for the output.

Links [5]

Table [10]

Solution: Type the following code in a note pad editor program and save it as example1.html.. open the file in a browser to see



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the output generated.

```
<html>
```

```
<body>
```

```
preceding content...
```

```
<a name="Stews">
```

```
<h3>Kumquat Stew Recipes</h3>
```

```
</a>
```

In the same or another

For more information on kumquats, visit our

```
<a href="http://www.kumquat.com/archive.html">
```

```
Kumquat Archive</a>,
```

and perhaps try one or two of our

```
<a href="http:
```

```
//www.kumquat.com/archive.html#Stews">
```



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Kumquat Stew Recipes.

```
<p><IMG SRC = "angel.jpeg">
```

```
<ul>
```

```
<li><a href="default.html">Home</a></li>
```

```
<li><a href="about/about.htm">About</a></li>
```

```
<li><a href="news/news.htm">News</a></li>
```

```
<li><a href="calendar/calendar.htm">Calendar</a></li>
```

```
<li><a href="photo_gallery/photo_gallery.htm">
```

```
Photo Gallery</a></li>
```

```
<li><a href="information_links/information_links.htm">
```

```
Links</a></li>
```

```
<li><a href="faq/faq.htm">FAQ</a></li>
```

```
<li><a href="contact/contact.htm">Contact</a></li>
```



```
</ul>
```

```
<table border cellpadding=5>
```

```
<caption align=bottom>
```

```
Kumquat versus a poked eye, by gender</caption>
```

```
<tr>
```

```
<td colspan=2 rowspan=2></td>
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```
<th colspan=2 align=center>Preference</th>
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</tr>
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<tr>
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<th>Eating Kumquats</th>
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<th rowspan=2>Gender</th>
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</tr>
<tr align=center>
<th>Female</th>
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<td>84%</td>
</tr>
</table>
</body>
</html>
```



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Learning Activities

Revision Questions

EXERCISE 1. ✍ With the use of code illustrate the rowspan and colspan attributes in html tables

EXERCISE 2. ✍ Identify and describe any three tags and elements used html code

EXERCISE 3. ✍ Which of these are elements and which are tags?<html>

```
<body>
```

```
preceding content...
```

```
<a name="Stews">
```

```
<h3>Kumquat Stew Recipes</h3>
```

```
</a>
```



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In the same or another

For more information on kumquats, visit our

`<a href="http://www.kumquat.com/archive.html">`

Kumquat Archive`</a>`,

and perhaps try one or two of our

`<a href="http://www.kumquat.com/archive.html#Stews">`

Kumquat Stew Recipes`</a>`.

`<p><IMG SRC = "angel.jpeg">`

`<ul>`

`<li><a href="default.html">Home</a></li>`

`<li><ahref="about/about.htm">About</a></li>`

`<li><ahref="news/news.htm">News</a></li>`

`<li><ahref="calendar/calendar.htm">Calendar</a></li>`



```
<li><a href="photo_gallery/photo_gallery.htm">
Photo Gallery</a></li>
<li><a href="information_links/information_links.htm">
Links</a></li>
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```

```
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```

```
</tr>
```

```
</table>
```

```
</body>
```

```
</html>
```

EXERCISE 4. 🖊 Describe the effect

ea<FRAMESET rows="30%,70%" cols="33%,34%,33%">
in a web browser window

EXERCISE 5. 🖊 Describe three attributes that have been used on the different tags we have seen so far and their effect on the behavior of these tags



References and Additional Reading Materials

- George Beekman,(2008) Tomorrow's Technology and You, Complete, 9/E,Oregon State University Ben Beekman, ISBN-10: 0135045045 • ISBN-13: 9780135045046 ©2010 • Prentice Hall • Paper, 672 pp
- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X
- Craig Grannell(2007 The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5



Solutions to Exercises

Exercise 1. Rowspan merges several rows depending on the integer value provided while colspan merges columns also depending on an integer value provided. Student should provide both a diagram and a code segment showing this

Exercise 1

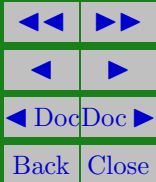


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Exercise 2. Put answer

Exercise 2



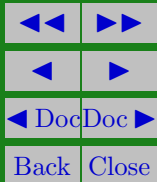


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Exercise 3. Put answer

Exercise 3





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Exercise 4. Put answer

Exercise 4

◀◀	▶▶
◀	▶
◀ Doc	Doc ▶
Back	Close



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Exercise 5. Put answer

Exercise 5

